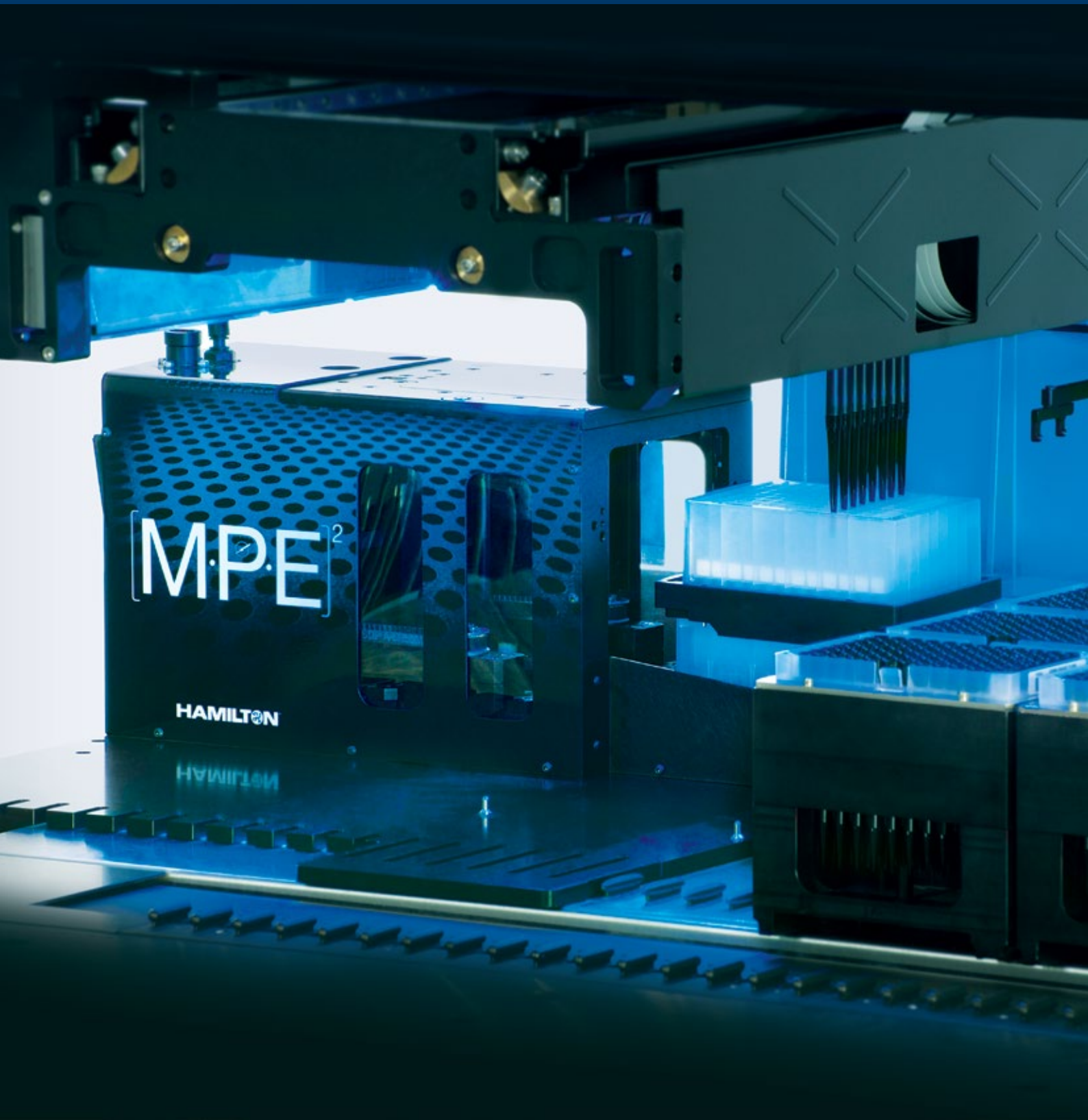


[MPE]²

Monitored Multi-Flow,
Positive Pressure Evaporative
Extraction] Module



Compact. Versatile. Affordable.



Accommodates filter collection plate/column heights up to 114 mm



Pipette into labware located on the [MPE]²



Maintains equal pressure across the filter plate, eliminating path of least resistance



Dual circuit, low and high pressure control up to 100 psi



The [Monitored Multi-flow, Positive Pressure Evaporative Extraction] module – [MPE]² – is the most versatile positive pressure module on the market. Providing fully automated, positive pressure extraction, reagent addition, and evaporation, the [MPE]² eliminates the need to purchase three separate instruments for these processes. With a small footprint, the [MPE]² is designed to easily integrate with the Microlab[®] NIMBUS[®], Microlab STAR[™], and Microlab VANTAGE Automated Liquid Handling System[®].

The [MPE]² offers interchangeable manifolds for 24-, 48-, and 96-well processing of plates, cartridges, and spin columns. The patented dual elevator design of the [MPE]² accommodates most filter, custom column, and collection plate combinations. Optimized features include a built-in waste tray that easily removes liquid waste processed from the filter plate, a plenum that evacuates aerosols when evaporating and drying down samples or tips in the tip drying station, and a reagent dispenser that can automatically fill plates or columns during equilibration, conditioning, and wash steps. The optional automated evaporator comes with a portable needle drying module that easily integrates with the [MPE]² to maximize walk-away time.

[MPE]² Hardware

- Interchangeable manifolds for 24-, 48-, and 96-well processing
- Optional evaporator: dry down samples with heated gas up to 140 °F
- Flexible gas supply: compressed air or nitrogen
- Optional reagent fill module: fast solvent addition with up to 15 reagent bottles
- Cap-mat sealer
- Tip drying station

This versatile device fulfills the automation needs of many labs, providing solutions for applications such as:

- Drug discovery: analytical chemistry, LC/MS
- Diagnostics: drugs of abuse
- Assay development: optimization of chemistries
- Forensic science: extractions
- Genomics: bead-based DNA extraction
- Proteomics: protein purification, extraction, buffer exchange, and desalting

Workflow

Example of a typical automated workflow using the [MPE]²



Condition/
Equilibrate



Sample
Addition



Wash



Elute Sample
to Collection



Evaporate



Seal with
Cap-mat

[MPE]² Specifications

Description	Specification		
Input Power (Primary)	Universal Supply: 100 – 240 VAC, 50 – 60 Hz, 2.5 A		
Output Power (Secondary)	Power: + 48 VDC ± 2%	Wattage: 221 Watts (maximum)	
Deck Positions	NIMBUS: 1 pedestal location	STAR: 7T Carrier (maximum)	Microlab VANTAGE: 7T Carrier (maximum)
Communication Type	Ethernet / USB		
CSA Certification	Installation Category: II	Pollution Degree: 2	
	Temperature	Relative Humidity	Altitude
Operating	5 °C (41 °F) to 40 °C (104 °F)	15% to 90% non-condensing	0 – 2000 m above sea level
Storage	-20 °C (-4 °F) to 70 °C (158 °F)	10% to 90% non-condensing	—
Physical Dimensions	[MPE] ² Logistics Module	[MPE] ² Control Box	Waste Bottle
Length	17.53 in (44.5 cm)	10.38 in (26.4 cm)	14.08 in (35.8 cm)
Width	6.25 in (15.9 cm)	8.43 in (21.4 cm)	5.78 in (14.7 cm)
Height	7.11 in (18.1 cm)	2.88 in (07.3 cm)	8.09 in (20.5 cm)
Weight	15.25 lbs (6.9 kg)	5 lbs (2.3 kg)	—

Filter / Collection Plate

Description	Specification	
Filter and Collection Plate with Spacer Combination	Maximum: 4.5 in (114.30 mm)	Minimum: None

Top Elevator Chemical Compatibility

Description	Specification
DMSO	Absolute / 100%
Sulfuric Acid	18 M / 36 N / ~96.0%
Sodium Hypochlorite	~10.0%
Halogenated Organics Including Chlorides	Absolute / 100%
Chloroform, Methylene Chloride	Absolute / 100%
Bromophenol Blue, Crystal Violet, Methylene Blue	Worst Case: 10.0% w/w (way above normal working concentration)
Acetonitrile	Absolute / 100%
Hexanes	Absolute / 100%
Alcohols	Absolute / 100%
Ethanol, Methanol, Isopropanol, 2-Propanol, etc.	Absolute / 100%
HCL	12.1 M / 12.1 N / ~37.0%
Acetic Acid, Glacial	17.4 M / 17.4 N / 99.7%
Formic Acid	23.6 M / 23.6 N / 90.5%
Hydrofluoric Acid	28.9 M / 28.9 N / 49.0%
Nitric Acid	15.8 M / 15.8 N / 70.0%
Perchloric Acid	11.7 M / 11.7 N / 70.0%
Phosphoric Acid	14.8 M / 44.6 N / 85.0%
Acetic Anhydride	97.0%
Ammonium Hydroxide	14.8 M / 14.8 N / 57.0%
Potassium Hydroxide	11.7 M / 11.7 N / 45.0%
Sodium Hydroxide	19.3 M / 19.3 N / 50.0%
Calcium Chloride	1 M
Sodium Bicarbonate	10.0%

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